

Hate Speech Detection Classifier

TF-IDF + Logistic Regression for binary hate-speech detection on the ETHOS dataset

Jashan Chaudhary - 25030422054 | Sanjeev Katragadda - 25030422059 | Agustya Gupta - 25030422011
BBA Artificial Intelligence • Symbiosis Artificial Intelligence Institute, Pune | Version 1.0 | October 2026

69.5%

Accuracy
majority baseline 56.5%

0.62

F1 (hate class)
precision 0.68 · recall 0.57

57.5%

Hate-class recall
50 of 87 caught

0.727

ROC AUC
macro F1 0.68 · test n=200

Model details

SCIKIT-LEARN

Task	Binary classification of English online comments (hate / non-hate)
Features	TF-IDF on 1-2 grams, lowercase, accents stripped, min_df 2, max_df 0.95, sublinear TF
Classifier	Logistic Regression, class_weight = balanced, max_iter 2000, seed 42
Pipeline	One scikit-learn Pipeline (TF-IDF -> LR); hand-set parameters, no tuning or model comparison
Artifacts	None saved; the model lives inside the notebook on Colab

Test-set performance

N = 200

METRIC	HATE (1)	NON-HATE (0)
Precision	0.676	0.706
Recall	0.575	0.788
F1-score	0.621	0.745
Support	87	113
Accuracy (overall)	0.695	

13 pts above the majority baseline; 37/87 hate comments missed; 24/113 non-hate comments falsely flagged.

Training & evaluation data

ETHOS • KAGGLE • PUBLIC

- ETHOS binary dataset: 998 online comments -> 998 after cleaning; 433 hate and 565 non-hate.
- Label rule: isHate >= 0.5 = hate (43.4%), else non-hate (56.6%). Median comment: 14 words.
- Split: stratified 80/20 -> 798 train / 200 test (87 hate, 113 non-hate). One split, no cross-validation.
- Audit slices: profanity vs no profanity and short vs long comments.

Intended use

IN SCOPE

- First-pass flagging of possible hate speech in English online comments for a human moderator.
- Teaching baseline for text classification and model auditing.

OUT OF SCOPE

- Automated removal, bans, or reports without human review.
- Non-English or code-mixed text, sarcasm, quoted hate, or text unlike ETHOS-style comments.
- Production use as it stands: it misses about 4 in 10 hate comments.

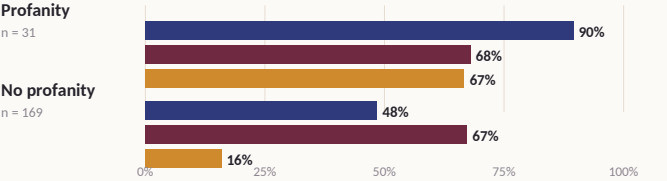
Audit 2 · short vs long comments

TEST SET

	ACCURACY	HATE RECALL	PRECISION	F1
Short (n = 104)	68.3%	59.5%	61.0%	60.2%
Long (n = 96)	70.8%	55.6%	75.8%	64.1%

Audit 1 · profanity vs no profanity

TEST SET



Hate recall: 89.5% with profanity vs 48.5% without. Non-hate false-positive rate: 67% vs 16%. Only 31 profanity comments, so treat as a warning.

Confusion matrices

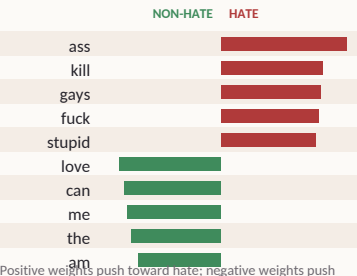
ROW % = SHARE OF ACTUAL CLASS

	All test comments		Profanity		No profanity	
	Non-hate	Hate	Non-hate	Hate	Non-hate	Hate
Non-hate	89 79%	24 21%	4 33%	8 67%	85 84%	16 16%
Hate	37 43%	50 57%	2 11%	17 89%	35 51%	33 49%

Subgroup matrices add up to the overall test-set matrix.

What drives predictions

WEIGHTS



Positive weights push toward hate; negative weights push toward non-hate. Hate-leaning terms include ass, kill, gays, fuck and stupid.

Limitations & risks

- Small data: 998 comments; only 200 in the test set.
- Misses hate: 37 of 87 hate comments were missed.
- Profanity shortcut: non-hate comments with profanity are falsely flagged much more often.
- Group words: identity terms can carry hate-leaning weights and cause benign false flags.
- Scope: English only; no sarcasm, negation scope or broader context.

Recommendations

- Use with human review: no automated removals, bans, or reports.
- Tune threshold: use a validation split if recall needs to increase.
- Expand the audit: test identity terms and more text styles.
- Improve data: remove near-duplicates and use cross-validation.
- Next: character n-grams, linear SVM, or BERT.